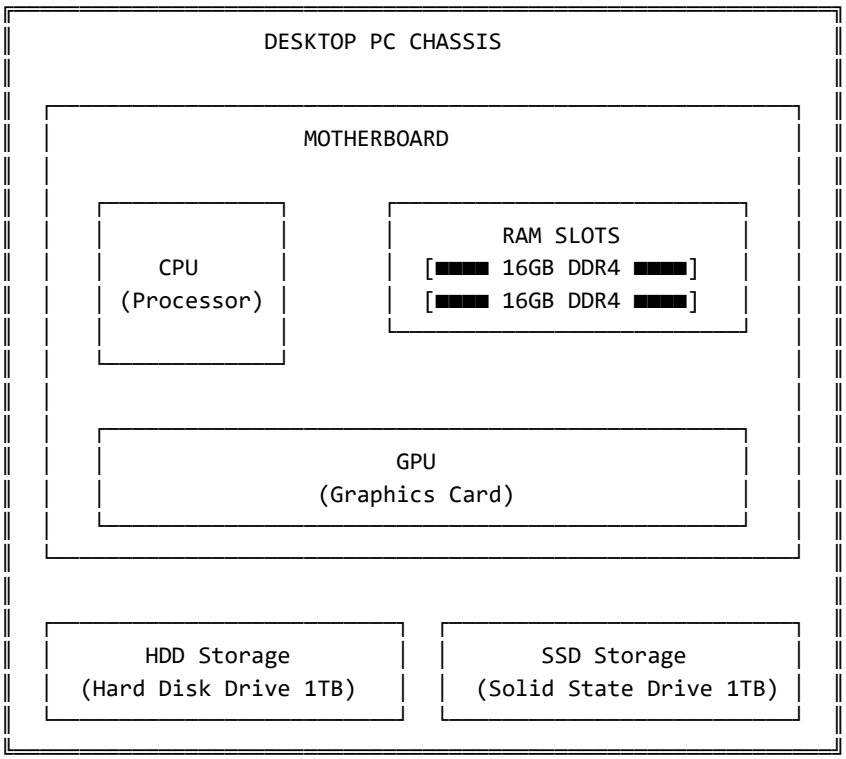
Hardware Components Assignment

Day 1: Overview & Component Analysis

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Task 1: Desktop PC Architecture Diagram

Below is a structural representation of a typical PC chassis, highlighting the primary hardware components and their placement:



Task 2: Storage Comparison (HDD vs SSD)

Storage Type	Speed	Durability / Shock Resistance	Typical Use Cases
HDD (Hard Disk Drive)	Slower (Mechanical parts, spinning platters)	Low (Fragile moving parts, prone to shock damage)	Mass storage, budget laptops/PCs, backup drives
SSD (Solid State Drive)	Much Faster (Flash memory)	High (No moving parts, highly shock-resistant)	OS drives, high-end laptops, gaming PCs

Task 3: Component Roles in Gaming (PUBG / Free Fire)

- **CPU (Central Processing Unit):** Acts as the brain of the computer, handling game logic, physics, and player inputs. It ensures actions like shooting or moving happen instantly.
- **RAM (Random Access Memory):** Stores active data the game needs immediately, such as map textures and character models. Sufficient RAM prevents stuttering when loading new areas.
- **GPU (Graphics Processing Unit):** Responsible for rendering visual elements, including lighting, shadows, and 3D models. A powerful GPU enables high resolutions and frame rates, crucial for spotting enemies.

Task 4: PC Upgrade Path for Latest FIFA

To run the latest version of FIFA smoothly, the following component upgrades are recommended, prioritized by impact:

1. **GPU (Graphics Card) -** Upgrading the GPU is critical for a sports game like FIFA. It is required for rendering highly detailed player models, stadium environments, and smooth animations at 60fps+ for optimal gameplay.

2. **SSD (Solid State Drive) -** Switching from an HDD to an SSD will significantly reduce load times between matches and menus, creating a seamless and snappy experience.

3. **RAM (Memory) -** Ensuring the system has at least 16GB of fast RAM prevents bottlenecks and stuttering, especially when loading dense crowds and dynamic weather effects.